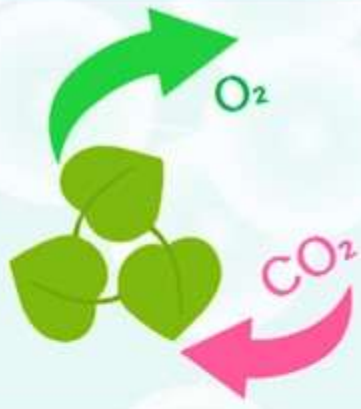




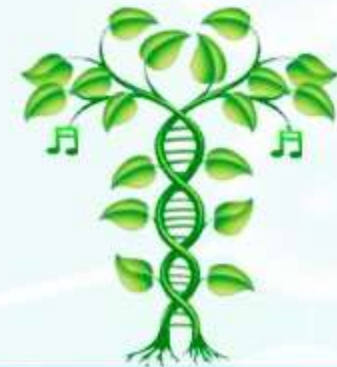
BIOLOGY



TYPES OF MACRONUTRIENTS



MODULE - 2





FOOD WHAT WE NEED ???

NUTRIENTS

DIETARY FIBRES

WATER



**MACRO -
NUTRIENTS**



**MICRO -
NUTRIENTS**



MAJOR NUTRIENTS IN OUR FOOD

MACRONUTRIENTS

- 1 CARBOHYDRATES
- 2 PROTEINS
- 3 FATS



MICRONUTRIENTS

- 1 VITAMINS
- 2 MINERALS





FOOD

WHAT WE NEED ???

NUTRIENTS

DIETARY FIBRES

WATER



**MACRO -
NUTRIENTS**



**MICRO -
NUTRIENTS**





MAJOR NUTRIENTS IN OUR FOOD

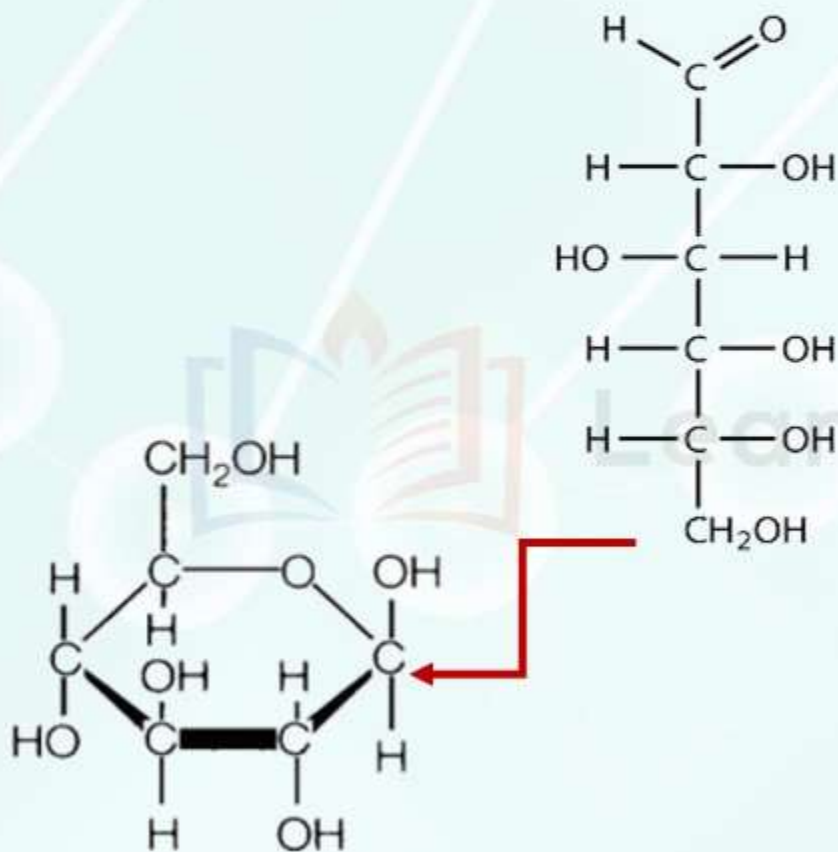
MACRONUTRIENTS

1 CARBOHYDRATES: PROVIDES ENERGY



MACRONUTRIENTS

WHAT IS CARBOHYDRATE MADE UP OF?



- 1 Carbohydrates are made of carbon, hydrogen and oxygen thus the name.
- 2 Hydrogen and oxygen in 2:1 ratio
- 3 They are also called saccharides



FORMS OF CARBOHYDRATES

MONOSACCHARIDE



GLUCOSE



FRUCTOSE



GALACTOSE

DISACCHARIDE



SUCROSE



MALTOSE



LACTOSE

POLYSACCHARIDE



STARCH
(in plants)



GLYCOGEN
(in animals)



CELLULOSE

SACCHARIDES MEANS SUGAR



FORMS OF CARBOHYDRATES A COMPARATIVE STUDY

1 MONOSACCHARIDES

- Simple sugars sweet in taste
- Soluble in water
- Needs no digestion straightaway absorbed in the body.



GLUCOSE

2 DISACCHARIDES

- Formed by two monosaccharide units
- They are soluble in water may taste sweet.
- They are broken down easily to absorbable units.



SUCROSE

3 POLYSACCHARIDES

- Formed by long chains of monosaccharide units.
- They are usually Insoluble in water.
- They are used as a storage component.



GLYCOGEN
(in animals)



FUNCTIONS OF CARBOHYDRATES



- 1 Carbohydrates are formed of carbon oxygen and hydrogen and oxidized in the cells to release energy.
- 2 Monosaccharide and disaccharide need no digestion and straightaway absorbed into the body.
- 3 Insoluble carbohydrates like starch and glycogen is used as storage in plants and animals respectively.
- 4 Cellulose makes cell wall in plants is another insoluble carbohydrate.
- 5 Carbohydrates are the principal sources of energy and an inseparable part of our diet.



MAJOR NUTRIENTS IN OUR FOOD

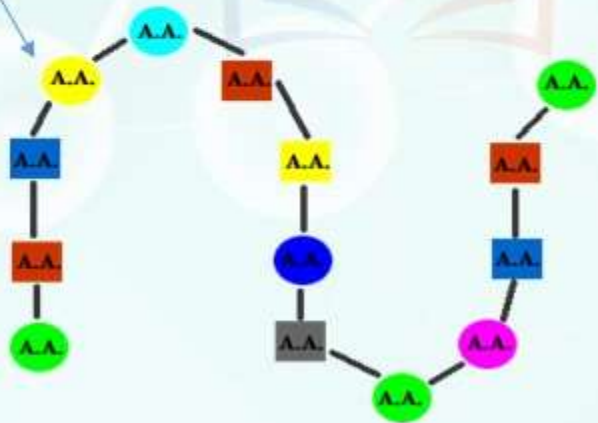
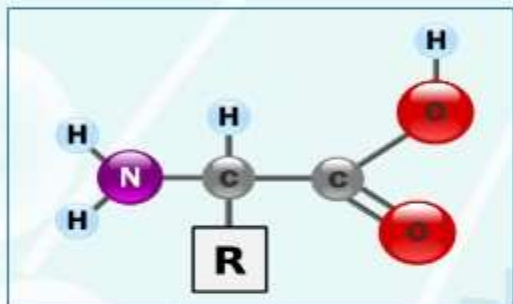
MACRONUTRIENTS

2 **PROTEINS:** BUILDS AND REPAIRS THE BODY



MAJOR NUTRIENTS IN OUR FOOD

WHAT IS PROTEIN MADE UP OF?



- 1 **AMINO ACID** are the organic compounds that is the primary units of protein.
- 2 There are 20 amino acids present in our body
- 3 Amino acids join together by a bond called as polypeptide bond.
- 4 Thus, by using peptide bonds amino acids join unit by unit and forms polypeptide chain.
- 5 Many such polypeptide chain is joined together and they coil up as a single structure called **PROTEIN**.



MACRONUTRIENTS

INDIAN COMMON SOURCES OF PROTEINS

VEGETARIAN



NON-VEGETARIAN





FUNCTIONS OF PROTEINS



- 1** Proteins are large chemical molecules that combine Carbon, hydrogen oxygen and nitrogen. Some contains sulphur and phosphorus.
- 2** Protein provide chemical material for the growth and repair of the body cells and tissues.
- 3** In time of emergency (starvation) proteins can be used by the body to release energy.
- 4** Deficiency of proteins leads to weakness and severe deficiency leads to deformation in the body(Kwashiorkor).
- 5** Thus, protein is another inseparable part of our diet required for our all body functions.





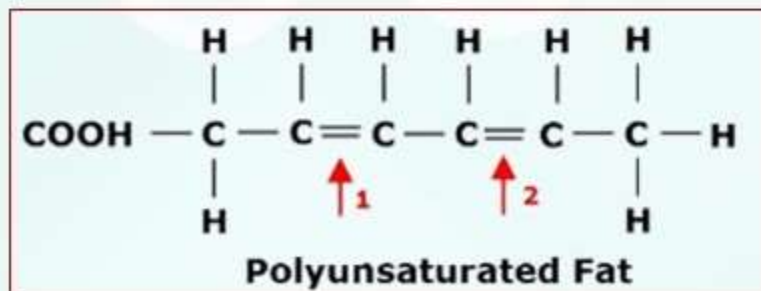
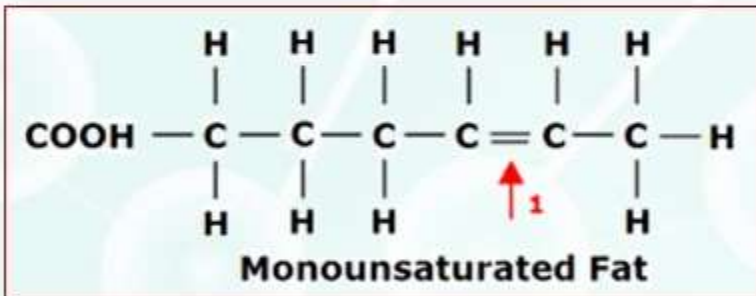
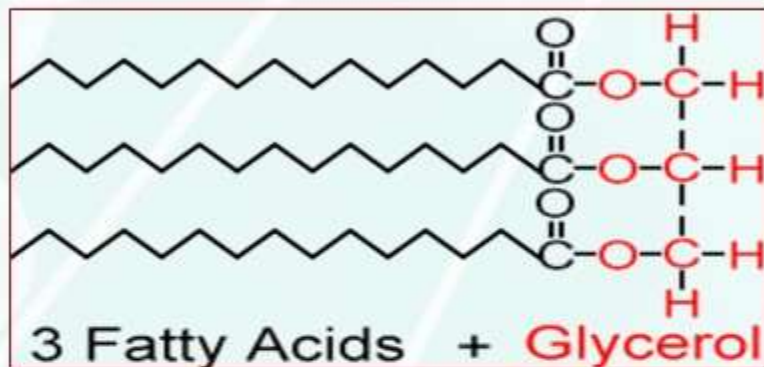
MAJOR NUTRIENTS IN OUR FOOD

MACRONUTRIENTS

3 **FATS**: PROVIDES LARGE AMOUNTS OF ENERGY



WHAT ARE FATS MADE UP OF?



- 1 Fats are triglycerides: esters of 3 fatty acid chains and glycerol.
- 2 Fats have unsaturated and saturated structure.
- 3 Arise from double bonds on the Carbon atom
- 4 It can have 1(mono) or >1(poly) unsaturation
- 5 Fatty acids that human require are called Essential Fatty Acid (EFA)



TYPES OF FATS WE GENERALLY EAT ...

HEALTHY FATS

UNSATURATED FATS



UNHEALTHY FATS

SATURATED FATS



FOODS CONTAINING HEALTHY FATS

MONOUNSATURATED FATS



POLYUNSATURATED FATS



FOODS CONTAINING BAD FATS

SATURATED FAT



TRANS FAT



Change of configuration in unsaturated fat during processing make it trans fat





ESSENTIAL FATTY ACID

OMEGA 3 FATTY ACID

Alpha-Linolenic Acid (ALA)

1. Chia seeds 2. flaxseed 3. hemp(cannabis)



OMEGA 6 FATTY ACID

Linoleic Acid (LA)

1. Safflower oil, 2. wheat-germ oil, 3. soybean oil



*The essential fatty acids are part of Poly unsaturated fatty acids
They help in improved brain function, digestion, good skin, hair nails, maintains good
immune system, hormones etc.*



FUNCTIONS OF FATS



- 1 They produce more energy than carbohydrates. If 1g carb gives 4 calorie whereas 1g fat gives 9.5 calorie
- 2 Mainly used by the body as a storage form of food for utilizing in harsh condition.
- 3 Fat serves as a solvent for fat soluble vitamins.
- 4 Body fat under the skin protects the body from low temperatures.
- 5 Thus, fats are an essential nutrient but it should be consumed in moderation.



DO ALL FOOD CONTAIN ALL NUTRIENTS?

TEST OF MACRONUTRIENTS

STARCH TEST TO IDENTIFY CARBOHYDRATES



IODINE



BREAD SLICE
(CONTAINING STARCH)



BREAD SLICE WILL
TURN BLUE BLACK

Starch reacts with iodine solution and turns blue black in colour, confirming carbohydrates in the food sample





DO ALL FOOD CONTAIN ALL NUTRIENTS?

TEST OF MACRONUTRIENTS

TEST TO IDENTIFY PROTEIN



SAMPLE SOLUTION



ADD COPPER SULPHATE
SOLUTION+ caustic soda



SAMPLE SOLUTION
TURNS VIOLET

Any protein containing food substance will react with basic copper sulphate solution and it will produce a violet colour



DO ALL FOOD CONTAIN ALL NUTRIENTS?

TEST OF MACRONUTRIENTS

TEST TO IDENTIFY FATS



SAMPLE FAT FOOD



PIECE OF PAPER RUB



PAPER TURNED
OILY AND TRANSLUCENT

Any oily patch on the paper shows that the paper contains fats





FOOD

NUTRIENTS

DIETARY FIBRES

WATER



 **MACRO -
NUTRIENTS**



**MICRO -
NUTRIENTS**



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